

popquiz 20 stellar parameters 4

⚠ This is a preview of the draft version of the quiz

Started: Mar 23 at 7:04pm

Quiz Instructions



Question 1 10 pts

The Algol binary system is called the



whale star



demon star



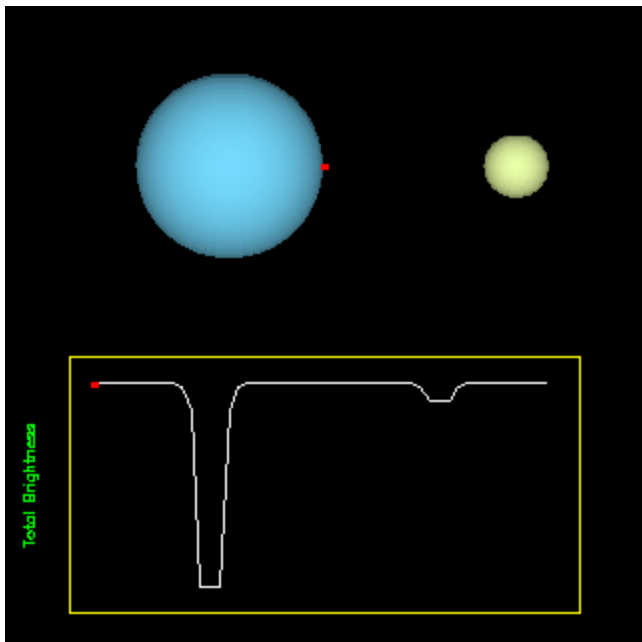
dog star



cepheid star



Question 2 10 pts



This is called



a spectroscopic binary



an optical binary



an eclipsing binary



a ghoul binary



Question 3 10 pts

When a star is on the **main sequence**, the luminosity depends mainly on the
(see slide 7.3)



temperature to the 2nd power



temperature to the 4th power



the temperature



to the size to the 2nd power



Question 4 10 pts

When a star leaves the main sequence,



it never leaves the main sequence



it runs out of helium in the core



it runs out of hydrogen in the core



it runs out of deuterium in the core



Question 5 10 pts

In a binary star system, both stars orbit their



a binary system is always static



the center of mass of the larger star



the center of the smaller star



common center of mass



Question 6 10 pts

cepheid variables have a special relationship between their luminosity and

☐

their period (time for a cycle)

☐

their rotation about their axis of rotation

☐

their color

☐

their apparent size in the sky



Question 7 10 pts

building on the previous question. Who discovered this relationship?

☐

Edwin Hubble

☐

Annie Jump Cannon

☐

Ejnar Hertzsprung

☐

henrietta Leavitt



Question 8 10 pts

When a star leaves the main sequence. Size matters. The luminosity is proportional to

☐

size to the fourth power

☐

size

☐

size to the third power

☐

size squared



Question 9 10 pts

In the HR diagram, below the main sequence (not part of it). In the lower part. we find the

☐

red giants

☐

blue giants

☐

red dwarfs

☐

white dwarfs



Question 10 10 pts



This illustration (7.4) shows an exotic binary system. It shows

- ☐ this is science fiction. never observed
- ☐ a red giant vampirizing a white dwarf
- ☐ a red giant connected to an accretion disk through a filament of dust
- ☐ a white dwarf vampirizing a red giant

Not saved

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